Assignment 2: Fields & Waves (ECE230), Winter 2024

*Total marks: 18. Deadline: 6 p.m. April 24, 2023. Submission: Dropbox outside B603, R&D Block (**soft copies submitted to google classroom will be ignored**)*

Plagiarism policy: ZERO tolerance towards copying assignments from others/ plagiarism from any other sources. Such cases will be dealt strictly according to the institute policy.

Late submission policy: -5/day after the submission deadline (starts immediately after 6 p.m. April 24. NO exceptions).

**Vector quantities must be represented with an overhead arrow, vector calculus operators must be written correctly, dot and cross products must be written properly. In case of any of these mistake -1 will be awarded per question.

NO credit will be given to the answers that do not accompany proper explanation.

Q1. In the lecture we have calculated the instantaneous Poynting vector as $\vec{\mathcal{S}} = \vec{\mathcal{E}} \times \vec{\mathcal{H}}$(i.e. the quantities indicated are time domain quantities). Consider the corresponding Fourier domain quantities as $\vec{E}$ and $\vec{H}$. Show that the time averaged Poynting vector is given by $Real\left(\vec{E} \times \vec{H}^*\right)$.

10 points

Q2. Consider the general form of the frequency dependent relative permittivity of a medium: $\epsilon_r(\omega) = 1 + \frac{Nq^2}{m\epsilon_0}\sum_j \frac{f_j}{\omega_j^2 - \omega^2 + i\omega\gamma_j}$

(a) How does the above expression get modified if we assume that all the electrons have same natural frequency of oscillation and same damping constants?

(b) Under certain assumptions, the relative permittivity of a medium can be written as: $\epsilon_r(\omega) = \left(1 - \frac{\omega_p^2}{\omega^2}\right)$. State the assumptions and derive the given form of $\epsilon_r(\omega)$.

$3 + (2 + 3) = 8$ points